

CS101

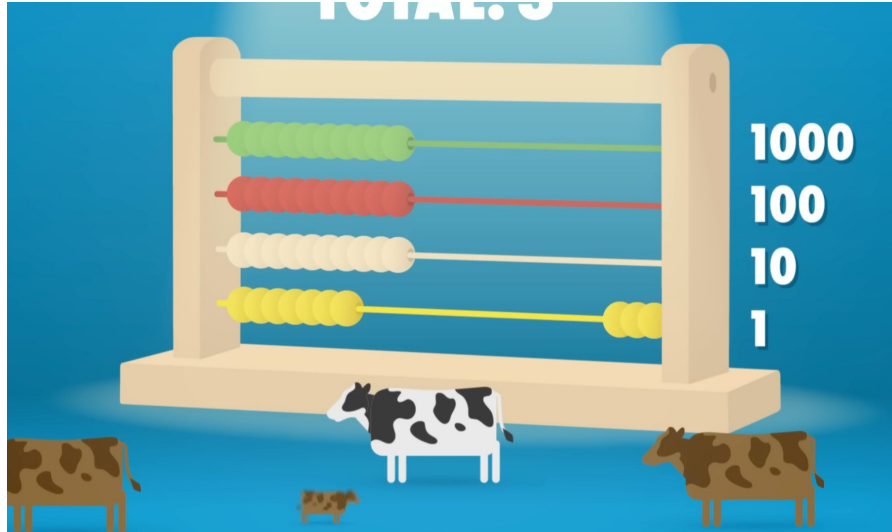
History of Computing

2/24

Agenda

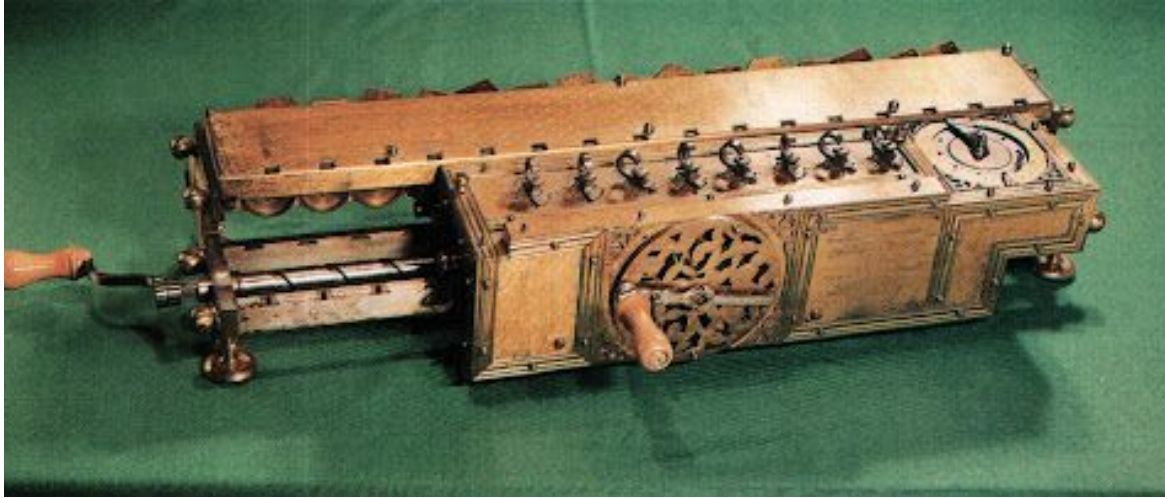
1. HW1 Competition
2. History of Computing (Videos/Slides/Kahoot)
3. Demo healthPrinter (if/else)
4. Quiz5 Practice
5. Windows OS Tips + Touch Typing

Early Computing: Crash Course Computer Science #1



Kahoot winner gets a fancy cookie!

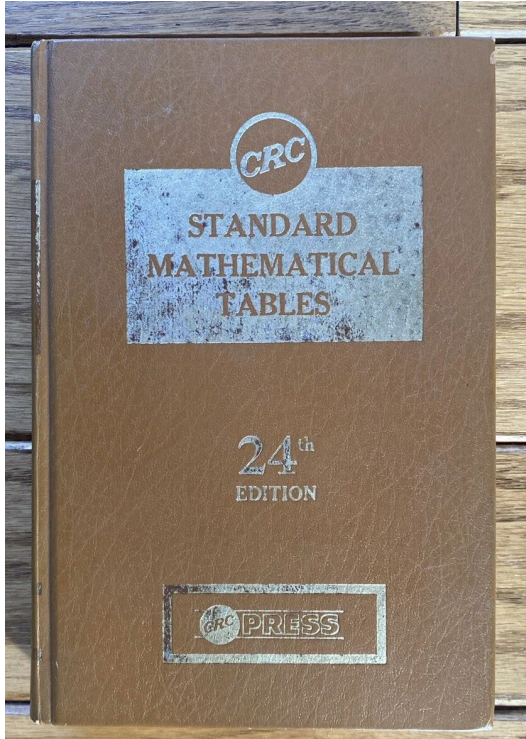
Stepped Reckoner



First device that could **add, subtract, multiply, divide**

- Idea **1672**
- Prototype **1673**
- Brass Version **1685**
- Technology used for **275 years** until electronic calculators

Mathematical Tables



Vintage Standard Mathematical Tables CRC 24th
Edition 1970's Math Tables

Table III. Four-Place Common Logarithms.

N	0	1	2	3	4	5	6	7	8	9
10	0000	0043	0086	0128	0170	0212	0253	0294	0334	0374
11	0414	0453	0492	0531	0569	0607	0645	0682	0719	0755
12	0792	0828	0864	0899	0934	0969	1004	1038	1072	1106
13	1139	1173	1206	1239	1271	1303	1335	1367	1399	1430
14	1461	1492	1523	1553	1584	1614	1644	1673	1703	1732
15	1761	1790	1818	1847	1875	1903	1931	1959	1987	2014
16	2041	2068	2095	2122	2148	2175	2201	2227	2253	2279
17	2304	2330	2355	2380	2405	2430	2455	2480	2504	2529
18	2553	2577	2601	2625	2648	2672	2695	2718	2742	2765
19	2788	2810	2833	2856	2878	2900	2923	2945	2967	2989
20	3010	3032	3054	3075	3096	3118	3139	3160	3181	3201
21	3222	3243	3263	3284	3304	3324	3345	3365	3385	3404
22	3424	3444	3464	3483	3502	3522	3541	3560	3579	3598
23	3617	3636	3655	3674	3692	3711	3729	3747	3766	3784
24	3802	3820	3838	3856	3874	3892	3909	3927	3945	3962
25	3979	3997	4014	4031	4048	4065	4082	4099	4116	4133
26	4150	4166	4183	4200	4216	4232	4249	4265	4281	4298
27	4314	4330	4346	4362	4378	4393	4409	4425	4440	4456
28	4472	4487	4502	4518	4533	4548	4564	4579	4594	4609
29	4624	4639	4654	4669	4683	4698	4713	4728	4742	4757
30	4771	4786	4800	4814	4829	4843	4857	4871	4886	4900
31	4914	4928	4942	4955	4969	4983	4997	5011	5024	5038
32	5051	5065	5079	5092	5105	5119	5132	5145	5159	5172
33	5185	5198	5211	5224	5237	5250	5263	5276	5289	5302
34	5315	5328	5340	5353	5366	5378	5391	5403	5416	5428
35	5441	5453	5465	5478	5490	5502	5514	5527	5539	5551
36	5563	5575	5587	5599	5611	5623	5635	5647	5658	5670
37	5682	5694	5705	5717	5729	5740	5752	5763	5775	5786
38	5798	5809	5821	5832	5843	5855	5866	5877	5888	5899
39	5911	5922	5933	5944	5955	5966	5977	5988	5999	6010
40	6021	6031	6042	6053	6064	6075	6085	6096	6107	6117
41	6128	6138	6149	6160	6170	6180	6191	6201	6212	6222
42	6232	6243	6253	6263	6274	6284	6294	6304	6314	6325
43	6335	6345	6355	6365	6375	6385	6395	6405	6415	6425
44	6435	6444	6454	6464	6474	6484	6493	6503	6513	6522
45	6532	6542	6551	6561	6571	6580	6590	6599	6609	6618
46	6628	6637	6646	6656	6665	6675	6684	6693	6702	6712
47	6721	6730	6739	6749	6758	6767	6776	6785	6794	6803
48	6812	6821	6830	6839	6848	6857	6866	6875	6884	6893
49	6902	6911	6920	6928	6937	6946	6955	6964	6972	6981
50	6990	6998	7007	7016	7024	7033	7042	7050	7059	7067
51	7076	7084	7093	7101	7110	7118	7126	7135	7143	7152
52	7160	7168	7177	7185	7193	7202	7210	7218	7226	7235
53	7243	7251	7259	7267	7275	7284	7292	7300	7308	7316
54	7324	7332	7340	7348	7356	7364	7372	7380	7388	7396

Reading the Table

The result is always a 4-digit decimal. For example, to find $\log(23.6)$:

1. Break your number into two parts: the first two digits (23) and the third digit (6)
2. Find row 23 in the N column
3. Find column 6 across the top
4. Read the value: 3730
5. So $\log(23.6) = 1.3730 \leftarrow$ you supply the integer part (called the *characteristic*)

Finding the Characteristic (the integer part)

The characteristic depends on where the decimal point is:

Number	Characteristic	Full log
236	2	2.3730
23.6	1	1.3730
2.36	0	0.3730
0.236	-1	-1 + .3730

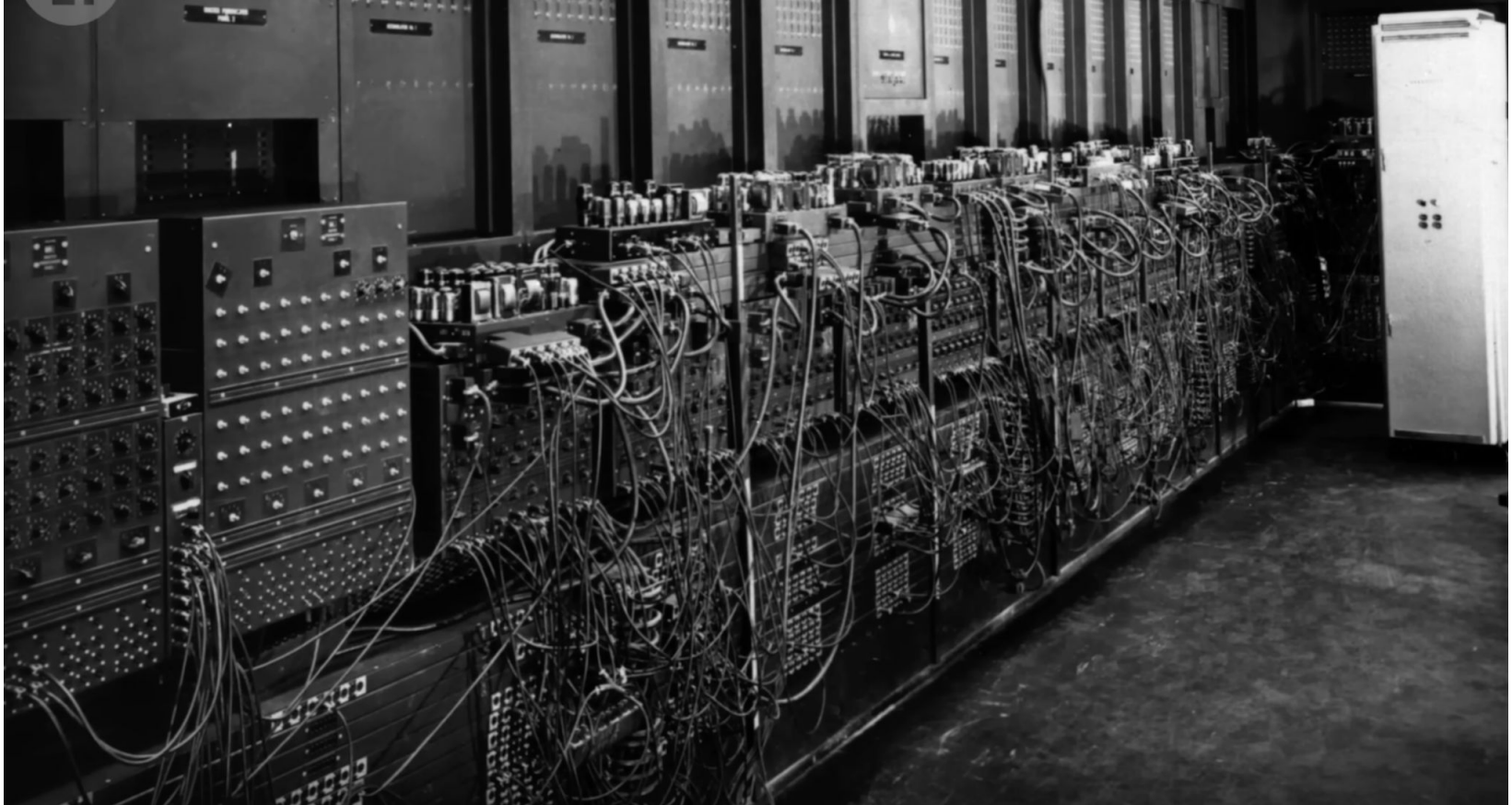
What is $\log(538)$

Ada Lovelace

The Story Behind The World's First Computer Programmer

- Envisioned computing machines that could **manipulate symbols**, not just numbers.
- **First algorithm:** step-by-step instructions for Bernoulli numbers
- Recognized machine could process music, graphics, text, etc (not just numbers)
- Thought of “**loops**” and “**functions**”

Electronic Computing: Crash Course Computer Science #2



ENIAC

1945

PHILADELPHIA, PA



18,000 VACUUM TUBES

Consumed 150 kW — enough to power 150 homes



30 TONS · 1,800 SQ FT

Filled an entire room at the University of Pennsylvania



5,000 OPERATIONS / SECOND

1,000× faster than any prior mechanical computer



SIX WOMEN PROGRAMMERS

Pioneering women engineers wrote all of ENIAC's programs

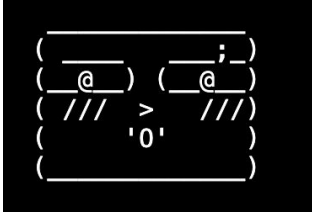
- ▶ First general-purpose electronic digital computer — one machine could solve any problem.
- ▶ Built for the U.S. Army in Philadelphia to calculate artillery ballistic tables in seconds, not days.
- ▶ Proved vacuum-tube electronics were reliable enough for large-scale computation.
- ▶ Developed at the Moore School of Electrical Engineering, University of Pennsylvania — operational 1945-1955.

"Mister Sandman" on a 1916 Behr Bros. Player Piano

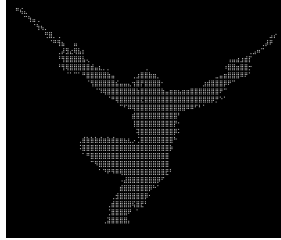


HW1 Competition! Select your favorite 2.

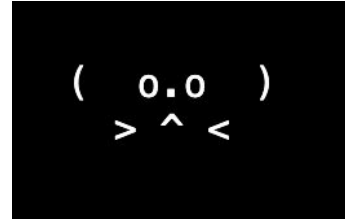
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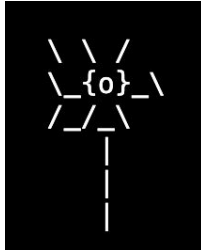
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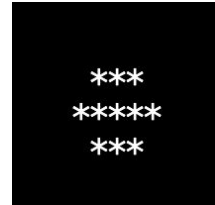
3



4



5



healthPrinter Demo

Quiz5 Kahoot

Announcements

1. Turn in HW2 !!
2. Bring Lab6 worksheet to turn in Thursday
3. Quiz5 on Thursday
4. Lab4 due Friday 2/27 at midnight